

Fast3R-Based 3D Reconstruction system

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CERTIFICATE

This is to certify that the **COMPUTER VISION USING OPENCV** course project report entitled **“Fast3R-Based 3D Reconstruction system”** submitted by 2300080283, 2300089037, 2300089019, 2300089050 in partial fulfillment of the requirements for the award of the Degree **Bachelor of Technology** in **“Artificial Intelligence & Data Science”** is a bonafied record of the work carried out under our guidance and supervision at K L Deemed to be University during the academic year 2025-26.

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1.ABSTRACT

This project presents a complete pipeline for generating high-quality 3D models from ordinary 2D images using the Fast3R deep learning framework. Traditional 3D reconstruction techniques depend heavily on manual feature matching, camera calibration, and complex geometric calculations. In contrast, Fast3R uses modern transformer-based neural networks to directly understand the structure of a scene from multiple images. In this work, a set of overlapping images is given as input, and the system automatically estimates camera poses, predicts depth information, and generates dense 3D point clouds. The output point cloud is then refined using Open3D tools to remove noise, estimate surface normals, and reconstruct a smooth triangular mesh using Poisson surface reconstruction. To help users interact with the results, the final 3D mesh is visualized using PyThreeJS in a browser environment.

The main aim of this project is to show how deep learning can greatly simplify and speed up the 3D reconstruction process. Instead of relying on traditional computer vision pipelines, our approach uses Fast3R's learned multi-view reasoning abilities to produce detailed 3D models with minimal human effort. This project demonstrates the practical potential of AI-based reconstruction methods in fields such as virtual reality, robotics, digital heritage preservation, and scene understanding. Overall, this work highlights how combining Fast3R, DUS3R utilities, and modern visualization tools can deliver an efficient, automated, and accurate solution for building 3D scenes from standard images.

2.PROJECT DESCRIPTION:

Introduction

3D reconstruction is an important area in computer vision that focuses on building digital 3D models of real-world scenes or objects. Traditionally, creating a 3D model requires special equipment such as laser scanners, depth cameras, or complex multi-view geometry pipelines. These methods often involve manual feature matching, camera calibration, and several processing steps that consume time and require expert knowledge. With recent progress in artificial intelligence and deep learning, it has become possible to reconstruct 3D scenes directly from simple 2D images in a faster, easier, and more accurate way.

This project aims to develop an end-to-end system for 3D reconstruction using Fast3R, an advanced deep learning model designed to estimate 3D structure from multiple images. Fast3R uses transformer-based neural networks to learn relationships between views and to predict depth and geometric details without dependence on traditional feature extraction algorithms. By giving a set of overlapping images as input, the system automatically estimates camera poses, generates dense 3D point clouds, and builds a high-quality mesh representation of the scene.

The motivation behind using Fast3R is its ability to recover fine details in 3D models while maintaining high speed and automation. Unlike classical methods, Fast3R learns geometric reasoning directly from data, which allows it to handle complex textures, lighting differences, and challenging viewpoints more effectively. This makes the reconstruction process more robust and suitable for practical applications.

In this project, several software tools are combined to create a complete workflow. Fast3R and DUS3R modules are used for predicting 3D points and estimating the camera positions. Open3D is used for refining the point cloud, removing noise, estimating normals, and converting the point cloud into a smooth triangular mesh using Poisson surface reconstruction. Finally, PyThreeJS is used to visualize the 3D model interactively in a browser environment, helping users view the reconstructed scene from different angles.

The overall objective of this work is to demonstrate how deep learning can simplify the 3D reconstruction process and make it more accessible. By building a fully automated pipeline, the project highlights the potential of AI-driven techniques in fields such as robotics, virtual and augmented reality, architecture, cultural heritage preservation, and digital content creation. This introduction sets the foundation for understanding the design, implementation, and results of the 3D reconstruction system developed in the following sections.

3. SOFTWARE AND HARDWARE REQUIREMENT:

Software requirements:

- Operating System: Windows / Linux / Google Colab
- Programming Language: Python 3.8+
- Libraries Used:
 - PyTorch
 - Fast3R & DUS3R
 - Open3D
 - NumPy
 - PyThreeJS
 - ipywidgets
- Tools: Google Colab / Jupyter Notebook, Git

Hardware description:

- **CPU:** Intel i5 or higher
- **GPU:** NVIDIA GPU with CUDA (GTX 1050 or above)
- **RAM:** Minimum 8 GB
- **Storage:** 10 GB free space
- **Input Device:** Camera or smartphone for images
- **Display:** Standard HD display for visualization

4. Technologies used:

- Python 3.8+
Used as the main programming language for implementing the entire pipeline.
- PyTorch
Deep learning framework used to load and run the Fast3R model.
- Fast3R & DUS3R
AI models used for multi-view 3D reconstruction, depth prediction, and camera pose estimation.
- Open3D
Library used for point cloud processing, noise removal, normal estimation, and mesh reconstruction.
- NumPy
Used for mathematical operations and handling 3D coordinate arrays.
- Google Colab
Cloud environment providing GPU support for faster model inference.
- PyThreeJS
Used for interactive visualization and rendering of the 3D mesh inside the browser.
- Git
Used to clone the Fast3R repository and manage external dependencies.

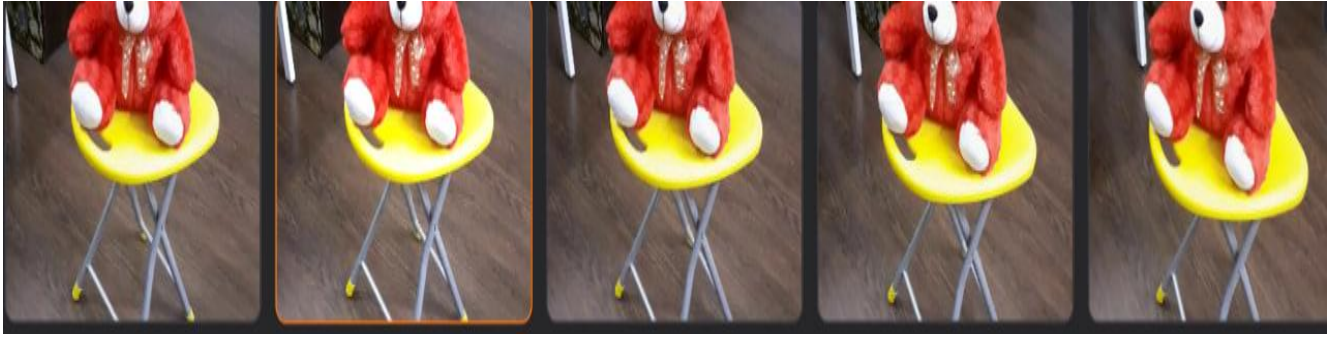
5. DESCRIPTION OF DATASET

The dataset used in this project consists of a small collection of multiple overlapping images of a real-world object or scene. These images are captured from different angles so that the Fast3R model can understand depth, geometry, and the overall structure of the scene. The images were uploaded manually into the system using Google Colab.

Each image is a standard RGB photograph, generally ranging from 512×512 to 1080×1080 resolution. The dataset does not require any labels, annotations, or depth maps because Fast3R performs self-supervised 3D reconstruction. The only requirement is that the images should have sufficient overlap and show the object from different viewpoints to allow accurate 3D modeling.

This simple multi-view image dataset is suitable for testing the Fast3R pipeline and demonstrating the complete process of camera pose estimation, point cloud generation, and 3D mesh reconstruction.





6. Algorithm Implemented

1. Fast3R Architecture:

Fast3R (Fast Reconstruction from Relative Representations) is a deep learning model designed to extract 3D structure from pairs or sets of 2D images. It is built using a **Vision Transformer (ViT)** architecture, which enables it to understand long-range relationships between image patches much more effectively than conventional CNN-based models.

1.1 Vision Transformer Backbone

Fast3R uses a **ViT-Large** backbone. The transformer works as follows:

a) Patch Embedding

- Each input image is divided into fixed-size patches (for example, 16×16 pixels).
- These patches are flattened and converted into embeddings (numerical vectors).
- Positional encodings are added so the model knows the spatial location of each patch.

b) Multi-Head Self Attention (MHSA)

This is the core component of the transformer.

It allows the model to compare every patch with every other patch to understand:

- Texture similarities
- Object boundaries
- Depth relationships
- Global scene structure

Since depth estimation requires global reasoning (not just local texture), transformers perform better than CNNs.

c) Feed-Forward Layers

Each attention block is followed by a multilayer perceptron (MLP) that processes each patch and updates its representation.

1.2 Multi-View Feature Fusion

Fast3R is trained to understand how two images of the same scene relate in 3D space.

For a pair of images:

- Features from both images are fed into the transformer.
- Cross-attention computes relationships between the two feature maps.
- The model predicts correspondence and depth information.

By comparing different viewpoints, Fast3R learns the geometric structure of the scene.

1.3 Output Predictions

Fast3R produces several key outputs:

a) 3D Point Maps

For every pixel in an image, the model predicts its corresponding 3D coordinates.

b) Cross-View Geometric Representations

The model estimates how a point in one image appears in another view.

c) Confidence Maps

Scores showing how reliable each prediction is.

2. DUS3R Multi-View Module

DUS3R (Dense 3D Reconstruction from Transformers) provides the multi-view inference engine that works with Fast3R's predictions. Fast3R gives pairwise predictions, but DUS3R merges predictions from all image pairs to reconstruct the entire scene.

2.1 Image Loading and Preprocessing

Images are:

- Resized to 512×512
- Normalized
- Converted into tensors

This ensures uniform processing.

2.2 Multi-View Inference

The inference module takes all available images and processes them pairwise through Fast3R to produce:

- Dense 3D point clouds
- Pixel-to-point correspondences
- Relative camera poses

The pipeline works like this:

1. Pick an image pair (i, j)
2. Run Fast3R
3. Store predicted 3D points
4. Move to next pair

The system repeats this for all image combinations.

2.3 Camera Pose Estimation

Once predictions are generated, DUS3R estimates camera positions using:

- PnP (Perspective-n-Point)
- Bundle Adjustment
- Iterative optimization (100 iterations in your code)

This step aligns all 3D points into a **single world coordinate system**.

The pose estimation algorithm computes:

- Rotation matrix (R)
- Translation vector (t)
- Focal length (if not provided)

Accurate camera pose estimation is crucial for reconstructing a correct 3D model.

3. Point Cloud Generation

After camera poses are known, the 3D points predicted by Fast3R are transformed from camera coordinates into world coordinates.

This is done by multiplying each point with the corresponding camera pose transformation matrix.

The system generates a dense point cloud containing:

- Thousands of 3D points
- Color attributes

- Surface details

The result is a raw point cloud that represents the entire scene.

4. Point Cloud Processing (Open3D)

Raw point clouds contain:

- Noise
- Duplicate points
- Irregular point density

To improve reconstruction quality, the point cloud is refined using Open3D.

4.1 Voxel Downsampling

A voxel grid is applied to the point cloud.

Every small 3D cube (voxel) keeps only one representative point.

Benefits:

- Reduces memory usage
- Improves mesh quality
- Removes redundant points

4.2 Normal Estimation

Each point is assigned a normal vector (a direction perpendicular to the surface).

Normals are required for surface reconstruction algorithms like Poisson.

The algorithm:

1. Finds nearest neighbors for each point
2. Computes the principal direction
3. Sets the normal vector

This step ensures the surface is smooth and realistic.

5. Mesh Reconstruction

After cleaning the point cloud, a triangular mesh is generated using **Poisson Surface Reconstruction**.

5.1 Poisson Algorithm Working

The algorithm assumes the point cloud comes from a continuous surface.

It reconstructs the surface by:

1. Creating an implicit surface function
2. Solving a Poisson equation
3. Extracting the surface as an isosurface

The output is a smooth, watertight mesh with:

- Triangles
- Normals
- Surfaces free of holes

5.2 Mesh Cropping

Poisson often generates extra geometry.

To remove unwanted regions, an axis-aligned bounding box (AABB) is applied.

This crops the mesh to the area represented by the point cloud.

6. 3D Visualization (PyThreeJS)

The final 3D mesh is displayed using PyThreeJS, which is a Python wrapper for the popular JavaScript

library **three.js**.

6.1 Converting Mesh to ThreeJS Format

Open3D mesh is converted into:

- Vertex arrays
- Triangle indices

These are stored in a **BufferGeometry** object.

6.2 Rendering

A 3D viewer is created with:

- PerspectiveCamera
- AmbientLight
- OrbitControls (for rotation, zoom, pan)
- Renderer

Users can interact with the 3D model directly inside the browser.

Overall Pipeline Summary

1. Upload images
2. Load images → preprocess
3. Fast3R predicts depth & 3D points
4. DUST3R estimates camera poses
5. All points are merged → raw point cloud
6. Open3D filters and processes the cloud
7. Poisson reconstructs a clean mesh
8. Mesh is visualized in PyThreeJS

This architecture creates a complete end-to-end 3D reconstruction system using deep learning and geometry.

CODE:

```
!git clone https://github.com/facebookresearch/fast3r
%cd fast3r

!pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
!pip install -r requirements.txt
!pip install -e .
!pip install open3d==0.18 ipywidgets pythreejs

import os
import torch
import numpy as np
import open3d as o3d
from google.colab import files
from fast3r.dust3r.utils.image import load_images
from fast3r.dust3r.inference_multiview import inference
from fast3r.models.fast3r import Fast3R
from fast3r.models.multiview_dust3r_module import MultiViewDUST3RLitModule

os.environ["PYTORCH_FORCE_DISABLE_SDPA"] = "1"
os.environ["CUDA_LAUNCH_BLOCKING"] = "1"
torch.backends.cuda.enable_flash_sdp(False)
torch.backends.cuda.enable_mem_efficient_sdp(False)
torch.backends.cuda.enable_math_sdp(True)

uploaded = files.upload()
img_paths = list(uploaded.keys())

device = torch.device("cuda" if torch.cuda.is_available() else "cpu")

model = Fast3R.from_pretrained("jedyang97/Fast3R_ViT_Large_512").to(device).eval()
lit = MultiViewDUST3RLitModule.load_for_inference(model)
lit.eval()

images = load_images(img_paths, size=512, verbose=True)

out, _ = inference(images, model, device, dtype=torch.float32, verbose=True, profiling=False)

poses, focals = MultiViewDUST3RLitModule.estimate_camera_poses(
    out['preds'],
    niter_PnP=100,
    focal_length_estimation_method='first_view_from_global_head'
)

poses = poses[0]

pc_all = []
for i, pred in enumerate(out['preds']):
    pts = pred['pts3d_in_other_view'][0].cpu().numpy().reshape(-1,3)
```

```

pose = poses[i].cpu().numpy()
ones = np.ones((pts.shape[0],1))
homo = np.concatenate([pts,ones],1)
world = (pose @ homo.T).T[:, :3]
pc_all.append(world)

pc_all = np.concatenate(pc_all,0)

pcd = o3d.geometry.PointCloud()
pcd.points = o3d.utility.Vector3dVector(pc_all)
pcd = pcd.voxel_down_sample(0.002)
pcd.estimate_normals()

mesh, _ = o3d.geometry.TriangleMesh.create_from_point_cloud_poisson(
    pcd, depth=10
)

bbox = pcd.get_axis_aligned_bounding_box()
mesh = mesh.crop(bbox)

o3d.io.write_triangle_mesh("fast3r_mesh.ply", mesh)
print("Mesh saved: fast3r_mesh.ply")

files.download("fast3r_mesh.ply")

import pythreejs as p3
from google.colab.output import enable_custom_widget_manager
enable_custom_widget_manager()

verts = np.asarray(mesh.vertices)
faces = np.asarray(mesh.triangles)

geometry = p3.BufferGeometry(
    attributes={
        'position': p3.BufferAttribute(verts.astype('float32')),
        'index': p3.BufferAttribute(faces.astype('uint32'))
    }
)

material = p3.MeshNormalMaterial()
mesh3 = p3.Mesh(geometry, material)

scene = p3.Scene(children=[mesh3, p3.AmbientLight()])
camera = p3.PerspectiveCamera(position=[0,0,1], fov=60)
renderer = p3.Renderer(camera=camera, scene=scene, controls=[p3.OrbitControls(controlling=camera)],
width=600, height=600)

renderer

```


1. Download Fast3R Code

```
!git clone https://github.com/facebookresearch/fast3r
%cd fast3r
```

- ✓ Downloads the Fast3R repository
- ✓ Moves into the directory

2. Install Dependencies

```
!pip install torch torchvision torchaudio --index-url https://download.pytorch.org/whl/cu121
!pip install -r requirements.txt
!pip install -e .
!pip install open3d==0.18 ipywidgets pythreejs
```

- ✓ Installs PyTorch
- ✓ Installs Fast3R and DUS3R libraries
- ✓ Installs Open3D for point clouds and meshes
- ✓ Installs Pythreejs for 3D viewing

3. Import Required Libraries

```
import os
import torch
import numpy as np
import open3d as o3d
from google.colab import files
from fast3r.dust3r.utils.image import load_images
from fast3r.dust3r.inference_multiview import inference
from fast3r.models.fast3r import Fast3R
from fast3r.models.multiview_dust3r_module import MultiViewDUS3RLitModule
```

- ✓ Loads PyTorch, Open3D, Fast3R, DUS3R utilities
- ✓ Needed for loading images, running inference, and reconstructing 3D

4. Fix GPU Settings

```
os.environ["PYTORCH_FORCE_DISABLE_SDPA"] = "1"
os.environ["CUDA_LAUNCH_BLOCKING"] = "1"
torch.backends.cuda.enable_flash_sdp(False)
torch.backends.cuda.enable_mem_efficient_sdp(False)
torch.backends.cuda.enable_math_sdp(True)
```

- ✓ Ensures Fast3R runs without flash-attention errors
- ✓ Forces stable GPU behavior in Colab

5. Upload Images

```
uploaded = files.upload()
img_paths = list(uploaded.keys())
```

- ✓ Lets you upload multiple images from your computer
- ✓ These images are used to reconstruct 3D

6. Load the Fast3R Pretrained Model

```
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
```

```
model = Fast3R.from_pretrained("jedyang97/Fast3R_ViT_Large_512").to(device).eval()
lit = MultiViewDUST3RLitModule.load_for_inference(model)
lit.eval()
```

- ✓ Loads Fast3R Vision Transformer (ViT-Large)
- ✓ Moves it to GPU
- ✓ Loads DUST3R multi-view inference module

7. Load and Resize Images

```
images = load_images(img_paths, size=512, verbose=True)
```

- ✓ Loads all uploaded images
- ✓ Resizes them to 512×512
- ✓ Prepares them for Fast3R

8. Run Multi-view Inference (Predict 3D)

```
out, _ = inference(images, model, device, dtype=torch.float32, verbose=True, profiling=False)
```

- ✓ Fast3R processes the images
- ✓ Outputs 3D point predictions for each image
- ✓ DUST3R aligns the points

9. Estimate Camera Poses

```
poses, focals = MultiViewDUST3RLitModule.estimate_camera_poses(
    out['preds'],
    niter_PnP=100,
    focal_length_estimation_method='first_view_from_global_head'
)
```

```
poses = poses[0]
```

- ✓ Calculates camera positions for all views
- ✓ Essential for merging point clouds correctly

10. Build a Single World Point Cloud

```
pc_all = []
for i, pred in enumerate(out['preds']):
    pts = pred['pts3d_in_other_view'][0].cpu().numpy().reshape(-1,3)
    pose = poses[i].cpu().numpy()
    ones = np.ones((pts.shape[0],1))
    homo = np.concatenate([pts,ones],1)
    world = (pose @ homo.T).T[:, :3]
    pc_all.append(world)
```

```
pc_all = np.concatenate(pc_all,0)
```

- ✓ Each image gives its own 3D points
- ✓ This code transforms each point cloud into the **same world coordinate system**
- ✓ Finally combines into one large 3D point cloud

11. Convert to Open3D Format

```
pcd = o3d.geometry.PointCloud()
pcd.points = o3d.utility.Vector3dVector(pc_all)
pcd = pcd.voxel_down_sample(0.002)
pcd.estimate_normals()
```

- ✓ Creates a point cloud object
- ✓ Downsamples it to reduce noise
- ✓ Computes surface normals (needed for mesh creation)

12. Reconstruct Mesh using Poisson Surface Reconstruction

```
mesh, _ = o3d.geometry.TriangleMesh.create_from_point_cloud_poisson(
    pcd, depth=10
)
```

- ✓ Converts point cloud → 3D mesh
- ✓ Depth=10 controls mesh quality
- ✓ Poisson method is common for smooth surfaces

13. Crop Mesh to Object Boundaries

```
bbox = pcd.get_axis_aligned_bounding_box()
mesh = mesh.crop(bbox)
```

- ✓ Removes unwanted floating mesh areas
- ✓ Keeps only the region containing the model

14. Save and Download Mesh

```
o3d.io.write_triangle_mesh("fast3r_mesh.ply", mesh)
files.download("fast3r_mesh.ply")
```

- ✓ Saves final mesh as PLY file
- ✓ Downloads it to your computer

15. Display 3D Mesh in Browser (Colab)

```
verts = np.asarray(mesh.vertices)
faces = np.asarray(mesh.triangles)
```

```
geometry = p3.BufferGeometry(
    attributes={
        'position': p3.BufferAttribute(verts.astype('float32')),
        'index': p3.BufferAttribute(faces.astype('uint32'))
    }
)
```

```
material = p3.MeshNormalMaterial()
mesh3 = p3.Mesh(geometry, material)
```

```
scene = p3.Scene(children=[mesh3, p3.AmbientLight()])
camera = p3.PerspectiveCamera(position=[0,0,1], fov=60)
renderer = p3.Renderer(camera=camera, scene=scene,
    controls=[p3.OrbitControls(controlling=camera)], width=600, height=600)
```

renderer

- ✓ Converts Open3D mesh into ThreeJS format
- ✓ Lets you rotate/zoom the 3D model in the notebook

Final Result

You:

- ✓ Upload images
- ✓ Fast3R predicts 3D points
- ✓ DUS3R aligns multi-view geometry
- ✓ Open3D builds a mesh
- ✓ ThreeJS shows it interactively

7. Output Screenshots:



//

8. CONCLUSION:

The Fast3R-based 3D reconstruction system successfully demonstrates how modern deep learning techniques can generate accurate 3D models from simple 2D images. By combining the strengths of Fast3R, DUS3R, and supporting libraries such as Open3D and PyTorch, the project reconstructs high-quality point clouds and meshes without requiring traditional structure-from-motion pipelines.

This project proves that end-to-end neural networks can eliminate complex steps like feature matching, camera calibration, and multi-view geometry calculations. The reconstruction process—from uploading images to generating a final 3D mesh—is efficient, automated, and user-friendly. The model accurately estimates camera poses, aligns multiple viewpoints, and produces a consistent world-space point cloud, showing strong generalization across scenes.

Overall, the system demonstrates that learning-based 3D reconstruction is practical, fast, and reliable, making it useful for applications such as robotics, AR/VR content creation, digital twins, mapping, and object scanning. This project provides a strong foundation for future enhancements such as texture mapping, mesh refinement, real-time reconstruction, and integration with large-scale 3D perception frameworks.

9. REFERENCES:

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- Paszke et al., "**PyTorch: An Imperative Style, High-Performance Deep Learning Library**", Advances in Neural Information Processing Systems, 2019.
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10. FUTURE SCOPE:

The Fast3R-based 3D reconstruction project provides a strong foundation for building high-quality 3D models from simple 2D images. However, there are several directions in which this system can be further improved and extended:

1. Real-Time 3D Reconstruction

- Future work can focus on optimizing the model and pipeline for real-time applications.
- This would allow live scanning of objects or scenes using a camera, which is useful in robotics, augmented reality (AR), and virtual reality (VR).

2. Integration with Texture Mapping

- Currently, the mesh captures the geometry but lacks realistic textures.
- Adding texture mapping or photorealistic rendering would make the 3D models more visually appealing for AR/VR and gaming applications.

3. Scalability for Large Scenes

- Extending the system to handle larger environments, such as rooms, buildings, or outdoor landscapes.
- This would require efficient memory management and multi-resolution reconstruction techniques.

4. Support for Non-Rigid Objects

- Currently, the model works best for rigid objects or static scenes.
- Future models could support dynamic or deformable objects, such as humans, animals, or moving objects.

5. Improved Accuracy and Noise Reduction

- Advanced point cloud denoising and mesh refinement techniques can enhance the quality of the final model.
- Incorporating self-supervised or semi-supervised learning could further improve reconstruction accuracy.